MUHAMMAD ZAMZURI BIN ZAMANI

Electronics Engineering Student (Analog & RF IC Design) ◇ Nationality: Malaysian

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OBJECTIVE

Seeking a **12–15 week internship starting 24 May 2027** as **[2027 Internship] Analog IC Design Intern** in MediaTek's **Radio System Solutions Division** (Singapore), to design and verify **RF analog building blocks** (oscillators/PLLs, LNAs, data converters) with hands-on CMOS design in **Cadence Virtuoso** and **Python/MATLAB** scripting.

EDUCATION

Master of Engineering (Diplôme d'Ingénieur) in Electronics **2024 - 2028**
Enseirb-Matmeca, Bordeaux INP *Talence, FR*

- **Analog/RF courses:** Integrated Circuits, **Feedback Amplifiers**, **Oscillators** (incl. HF/LC), **Voltage References**, Analog & Digital Modulations, Signal Conditioning
- **Digital courses: VHDL Synthesis & FPGA Design**, Digital Circuit Technologies, 32-bit MCUs, DSP

Bachelor (B.U.T.) of Electrical Engineering and Industrial Computing **2022 - 2024**
*IUT Nice Côte d'Azur, Université Côte d'Azur — Electrical and Energy Management, equivalent **3.5 CGPA*** *Nice, FR*

Foundation in Engineering (French Engineering Preparatory Programme) **2021 - 2022**
UniKL MFI (Malaysia France Institute) *Kuala Lumpur, MY*

ANALOG / RF IC DESIGN PROJECTS

Negative-Resistance LC Oscillator in 350 nm CMOS **2026**

- Designed a **cross-coupled NMOS** oscillator whose **negative resistance** ($-2/g_m$) cancels the losses of two 10 nH ($Q_L = 20$) inductors with 6.76 pF ($R_T \approx 544\,\Omega$), biased at 2.2 mA from 3.3 V: **433 MHz**, **1.75 V** differential amplitude and **−118.8 dBc/Hz at 100 kHz** in PSS/pnoise, meeting the >1 V_{pp}, < −110 dBc/Hz and <5 mA specifications.
- Sized devices by hand ($g_m = 2\times$ start-up value); verified the tank (S-param), start-up, amplitude and **phase noise** with **tran, PSS and pnoise** in **Cadence Virtuoso (Spectre)** against theory.

4-bit Flash ADC in 0.16 μm CMOS **2025**

- Designed the **comparator array** and calibrated resistor ladder of a **data converter** and validated the architecture in **Cadence Virtuoso**: **206 mV LSB** on a 3.3 V reference.

Optoelectronic FSK Data Link **2026**

- Built a **photodiode transimpedance front-end** with **compensated feedback**, band-pass filtering and envelope detection that recovers a 1200 baud bitstream over 30 cm of free space under ambient room lighting, on a single 5 V rail.

4-bit R-2R DAC with Sample-and-Hold Deglitching **2026**

- **SPICE** transient analysis; removed a **12 mV op-amp overshoot**, masked carry glitches with a **20 μs** S&H clock.

RESEARCH EXPERIENCE

R&D Engineer - INTERNSHIP **Jun 2026 - Aug 2026**
Graduate Student Researcher / R&D Intern **Jun 2025 - Aug 2025**
LDMRC, Department of Physics, Universiti Malaya *Kuala Lumpur, MY*

- Wrote **Python scripts** to generate **interdigitated electrode (IDE)** geometries parametrically (0.05 mm gap, 0.1 mm finger width), routed them in **KiCad**, fabricated the electrodes, and deposited and annealed PEDOT:PSS films to produce capacitive humidity sensors, leading to a **co-authored publication**.

R&D Hardware/Software - INTERNSHIP **Feb 2024 - Apr 2024**
LEAT, CNRS – Université Côte d'Azur *Biot, FR*

- Designed and routed the control board for a reconfigurable **Planar Inverted-F Antenna (PIFA)** for **RF/IoT** links, and wrote **STM32** firmware in **C** to switch the antenna between frequency bands.

PUBLICATION

M. Qianhui, M. Alserhan, **Z. Zamani**, *et al.*, "Humidity-induced capacitance as a probe of formulation-specific phase separation in PEDOT:PSS thin films," *Surfaces and Interfaces*, vol. 98, art. 110567, 2026. DOI: 10.1016/j.surfin.2026.110567

SKILLS

Analog / RF	CMOS transistor sizing, LC oscillators, phase noise, TIA front-ends, data converters (ADC/DAC)
EDA Tools	Cadence Virtuoso (Spectre: AC, tran, S-param, PSS, pnoise), SPICE, ADS, MATLAB, KiCad
Scripting / Digital	Python, MATLAB, C, C++; VHDL, Verilog, Vivado, ModelSim, Quartus; STM32, ESP32, PIC
Languages	Malay (**Native**), English (**Bilingual**), French (**Fluent**), Thai (**Basic**), Mandarin (**Basic**)

LEADERSHIP

- **Project Manager**, Malaysian Students' Association in France (2023–26): led **"From Strategy to Silicon"**, a semiconductor industry forum for **100+** students; co-founded a mentorship programme (**150+**).